

AEM-X100 — Startup and Shutdown Procedure

Equipment: AEM-X100 Wafer Handling System

Document ID: SYN-X100-STARTUP_SHUTDOWN_PROCEDURE **Revision:** A **Status:** Synthetic Training Data

IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Define a fictional normal startup and controlled shutdown sequence for retrieval and workflow testing.

1. Preconditions

- Only a trained fictional service engineer/operator should perform the procedure.
- Confirm the work area is clear and no maintenance activity is in progress.
- Confirm the emergency stop is released and all required safety interlocks report normal.
- Confirm the AEM-X100 is connected to its normal utility interfaces defined by the site procedure.

2. Startup Sequence

- Step 1 — Perform the pre-start visual check.
- Step 2 — Confirm the SCU displays SAFE/READY conditions without active safety alarms.
- Step 3 — Initiate system power-up using the normal operator control.
- Step 4 — Wait for the SCU to complete self-checks.
- Step 5 — Verify WPS, ALM, WTM and VHM status are reported as available.
- Step 6 — Run the fictional homing sequence from the normal operator interface.
- Step 7 — Confirm system state changes to READY.
- Step 8 — If a fault remains active, stop the startup sequence and refer to Error Code Reference and Escalation Procedure.

3. Controlled Shutdown

- Step 1 — Complete or safely stop the current wafer transfer.
- Step 2 — Confirm no wafer is left in an intermediate position.
- Step 3 — Place the system into the normal IDLE state.
- Step 4 — Initiate shutdown from the SCU.
- Step 5 — Verify the SCU indicates the system is safely stopped.
- Step 6 — Do not open protected access areas unless the applicable safety procedure has been satisfied.

4. Troubleshooting Cross-References

- If the system does not reach READY, search Error Code Reference for the displayed alarm.
- If WPS does not report correctly, search Sensor Troubleshooting.
- If an interlock remains active, search Safety Procedure and Escalation Procedure.
- If the required procedure cannot be confirmed from approved documentation, escalate rather than improvise.

End of synthetic document.